

3rd Mar 2026 / DA6360

⇒ Partial Derivatives

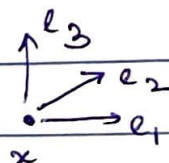
$$\frac{\partial f}{\partial x_i}(x_1, \dots, x_d) =$$

$$\lim_{\delta \rightarrow 0} \frac{f(x_1, \dots, x_d) - f(x_1, \dots, x_d)}{\delta}$$

$$= \lim_{\delta \rightarrow 0} \frac{f(x + \delta e_i) - f(x)}{\delta}, \quad e_i = \begin{bmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{bmatrix} \leftarrow \begin{matrix} i^{\text{th}} \\ \text{coord.} \end{matrix}$$

- simple approx. of only i^{th} coordinate

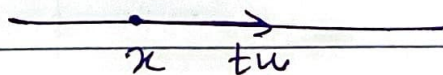
$$f(x + \delta e_i) \approx f(x) + \delta \frac{\partial f}{\partial x_i}$$



- but what happens for a small step in a general "direcⁿ" $u \in \mathbb{R}^d$?
(Directional Derivative)

$\leftarrow h(t) = f(x + tu)$

^{1D} funcⁿ



$$\frac{dh(t)}{dt} = \sum_{i=1}^d \frac{\partial f}{\partial x_i} u_i$$

gradient of the funcⁿ at a pt. x is

$$\nabla f(x) = \begin{bmatrix} \frac{\partial f}{\partial x_1} \Big|_x \\ \vdots \\ \frac{\partial f}{\partial x_d} \Big|_x \end{bmatrix}$$

$\Big|_x$
means
evaluated
at x .

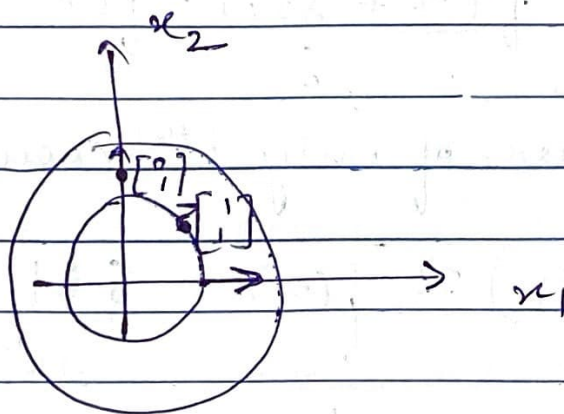
$$\frac{dh(t)}{dt} = \langle \nabla f(x), u \rangle$$

⇒ Gradient Field

$$\nabla f: \mathbb{R}^d \rightarrow \mathbb{R}^d$$

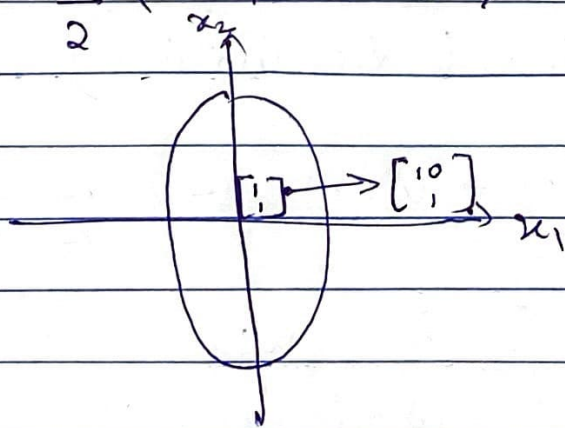
$$f(x) = \frac{1}{2}(x_1^2 + x_2^2)$$

$$\nabla f(x) = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$



$$f(x) = \frac{1}{2}(10x_1^2 + x_2^2)$$

$$\nabla f(x) = \begin{bmatrix} 10x_1 \\ x_2 \end{bmatrix}$$



Quiver Plots
(Matplotlib)

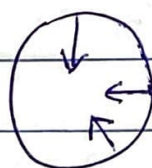
⇒ Gradient Descent in d-dim

$$x_{t+1} = x_t - \alpha \frac{df}{dx}$$

$$= x_t - \alpha f'(x_t)$$

$$x_{t+1} = x_t - \alpha \nabla f(x_t)$$

negative gradient field



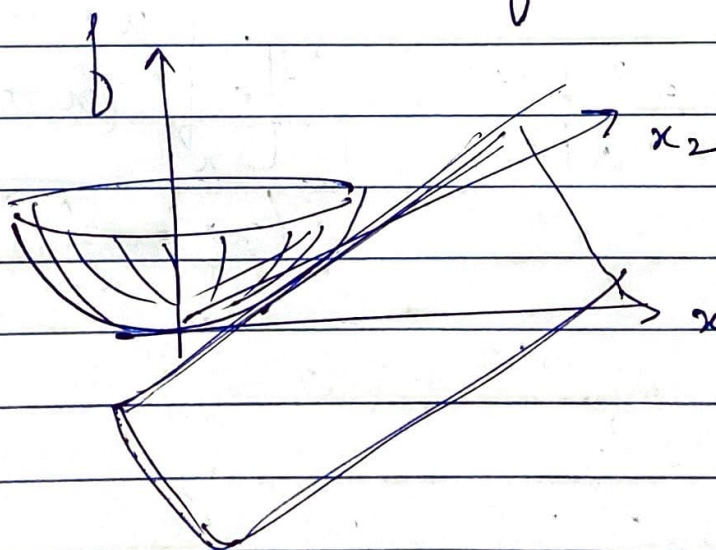
- First Order Condiⁿ

$$f(\hat{x} + t(x - \hat{x})) = h(t)$$

$$h(t) \approx h(0) + h'(0) \underline{t}$$

$$h(1) \approx h(0) + h'(0) \underline{1}$$

$$f(\hat{x} + t(x - \hat{x})) = f(\hat{x}) + \underbrace{\langle \nabla f(\hat{x}), t(x - \hat{x}) \rangle}_{\text{provides a tangent plane.}}$$



graph of the entire funcⁿ gives us a tangent plane.

- Remainder version

$$h(t) = h(0) + h'(z_t) \frac{t}{1!} \quad z_t = (0, t)$$

$$h(1) = h(0) + h'(z_1) \quad z_1 = (0, 1)$$

$$f(x) = f(\hat{x}) + \langle \nabla f(\bar{x}), (x - \hat{x}) \rangle$$

$$\bar{x} = \hat{x} + z(x - \hat{x})$$

- Second Order Approximation

$$h(t) = f(\hat{x} + t(x - \hat{x}))$$

$$h(t) = h(0) + h'(0)t + \frac{h''(0)t^2}{2!}$$

$$h(1) = h(0) + h'(0) + \frac{h''(0)}{2}$$

$$h'(t) = \frac{dh(t)}{dt} = \sum_{i=1}^d \frac{\partial f}{\partial x_i} (\hat{x} + t(x - \hat{x}))$$

$$h''(t) = \frac{d^2 h(t)}{dt^2} = \sum_{i=1}^d \frac{d}{dt} \left[\frac{\partial f}{\partial x_i} (\hat{x} + t(x - \hat{x})) \right]$$

$$g_i(t) = \frac{\partial f}{\partial x_i} (\hat{x} + t(x - \hat{x})) \quad g_i(t)$$

$$\frac{d}{dt} g_i(t) = \sum_{j=1}^d \frac{\partial g_i}{\partial x_j} (x_j - \hat{x}_j)$$

$$h''(t) = \sum_{i=1}^d \left[\sum_{j=1}^d \frac{\partial g_i}{\partial x_j} (x_j - \hat{x}_j) \right] (x_i - \hat{x}_i)$$

$$g_i = \frac{\partial f}{\partial x_i}$$

$$h''(t) = \sum_{i=1}^d \left[\sum_{j=1}^d \left(\frac{\partial}{\partial x_j} \frac{\partial f}{\partial x_i} (\hat{x} + t(x - \hat{x})) \right) (x_j - \hat{x}_j) \right] (x_i - \hat{x}_i)$$

$$h''(t) = \sum_{i,j=1}^d \frac{\partial^2 f}{\partial x_j \partial x_i} (\hat{x} + t(x - \hat{x})) (x_i - \hat{x}_i) (x_j - \hat{x}_j)$$

rewrite
in
matrix
form

$$f(x) \approx f(\hat{x}) + \langle \nabla f(\hat{x}), (x - \hat{x}) \rangle + \frac{1}{2} (x - \hat{x})^T \nabla^2 f(\hat{x}) (x - \hat{x})$$

- FONC

Let ∇f be a continuous funcⁿ.

$x_* \in \mathbb{R}^d$ is a local minima $\Rightarrow \nabla f(x_*) = 0$

suppose not $\nabla f(x_*) \neq 0$

$$u = -\delta \operatorname{sign}(\nabla f(x_*))$$

$$f(x) = f(x_*) + \langle \nabla f(\bar{x}), u \rangle$$

$$= f(x_*) + \sum_{i=1}^d \frac{\partial f}{\partial x_i}(\bar{x}) \text{sign} \frac{\partial f}{\partial x_i} x_*$$

- SOSC

Let $\nabla f, \nabla^2 f$ be conti., if x_* is a local minimizer

$$\rightarrow \nabla f(x_*) = 0$$

$$\rightarrow x^T \nabla^2 f(x) x \geq 0 \quad (\text{positive semi-def. hessian matrix})$$

$$\nabla^2 f = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\text{or } \nabla^2 f = \begin{bmatrix} 10 & 0 \\ 0 & 0.1 \end{bmatrix}$$

$$\text{or } \nabla^2 f = \begin{bmatrix} 1 & 0 \\ 0 & -0.1 \end{bmatrix}$$

suppose not $\Rightarrow$ for some u , $u^T \nabla^2 f u < 0$

$$f(x) + \langle \nabla f(x_*), x - x_* \rangle + (x - x_*)^T \nabla^2 f(x) (x - x_*)$$